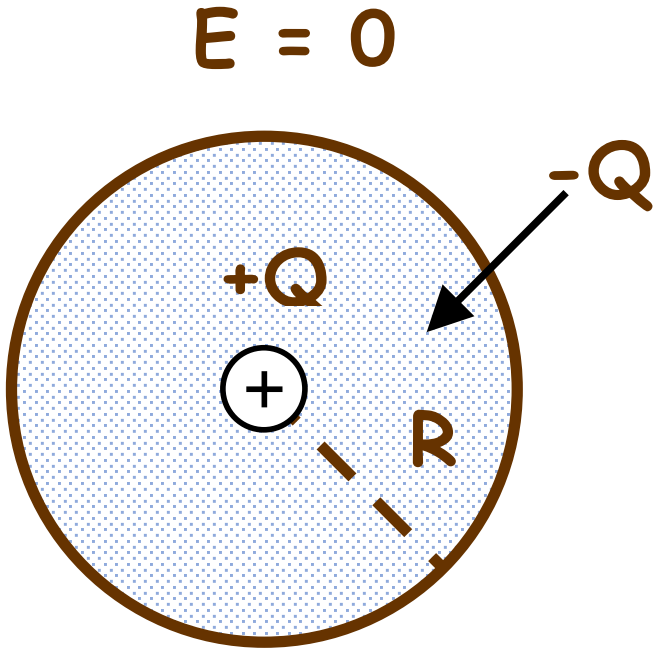
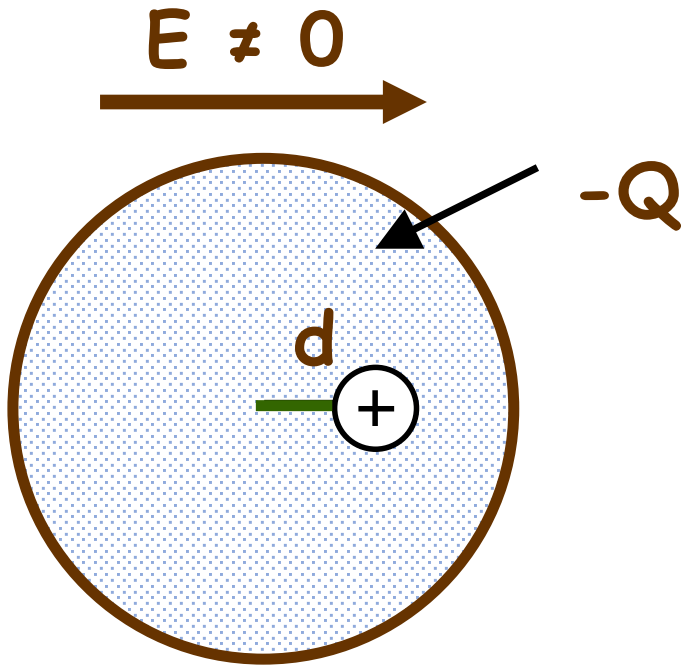
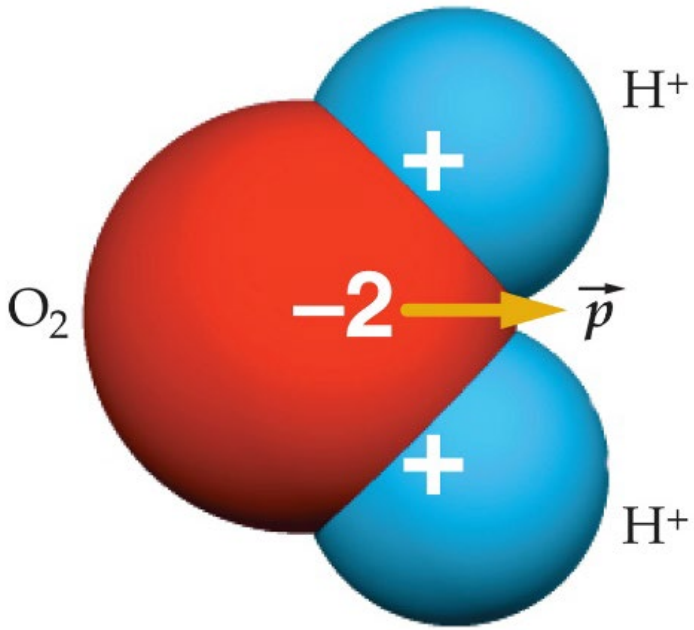
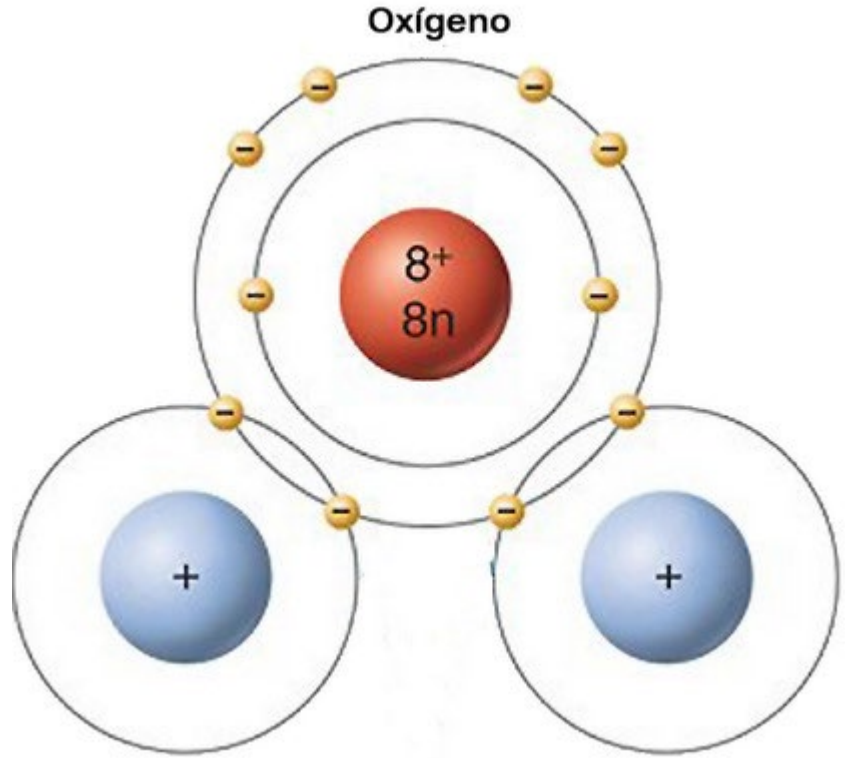


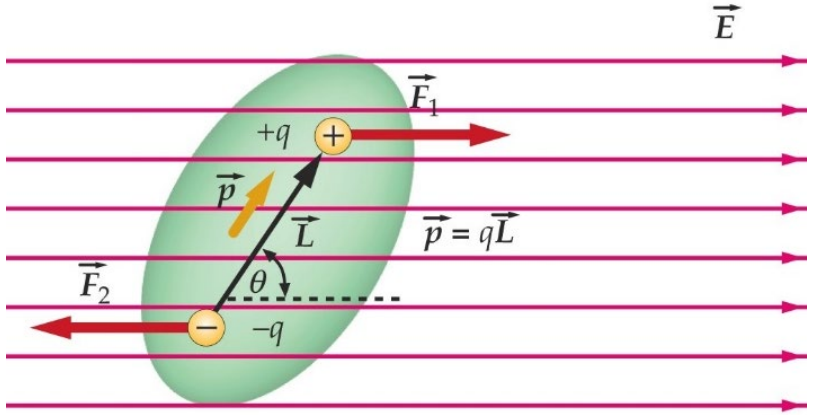
Átomo sin polarizar



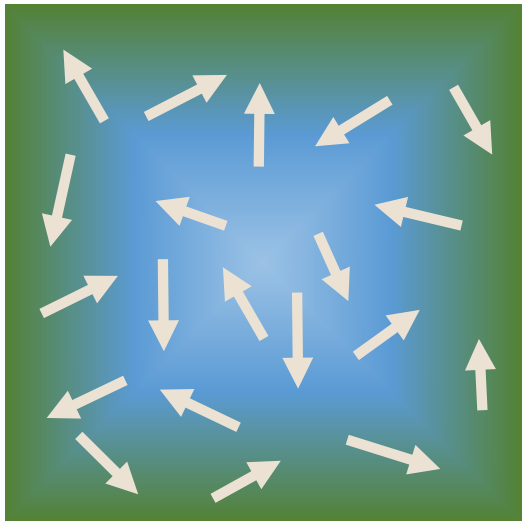
Átomo polarizado


H+
O2
−2
p
H+

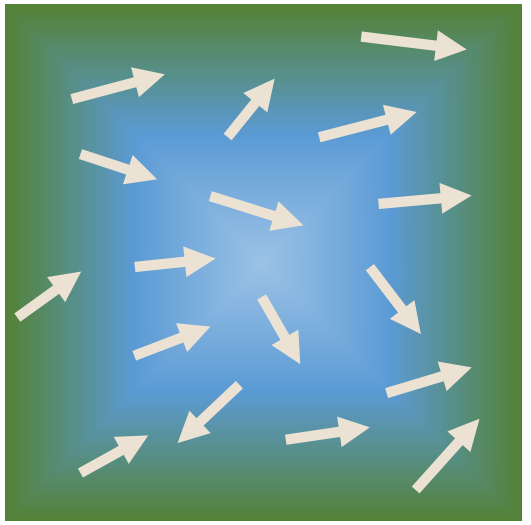

Oxígeno
8+
8n

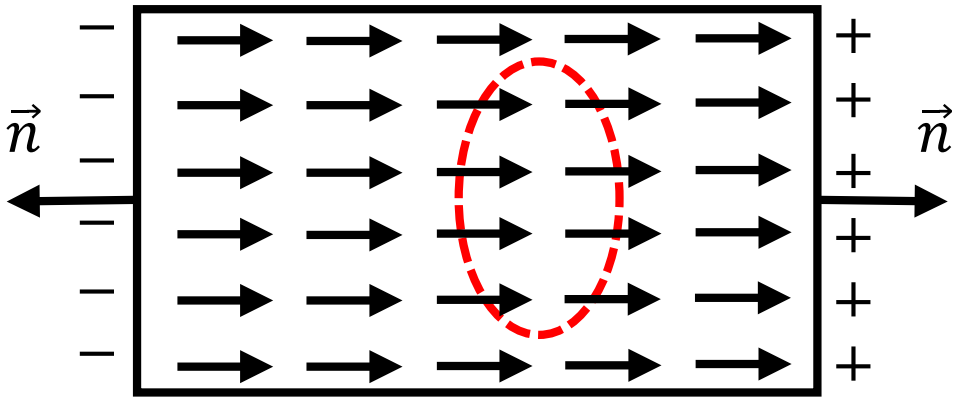


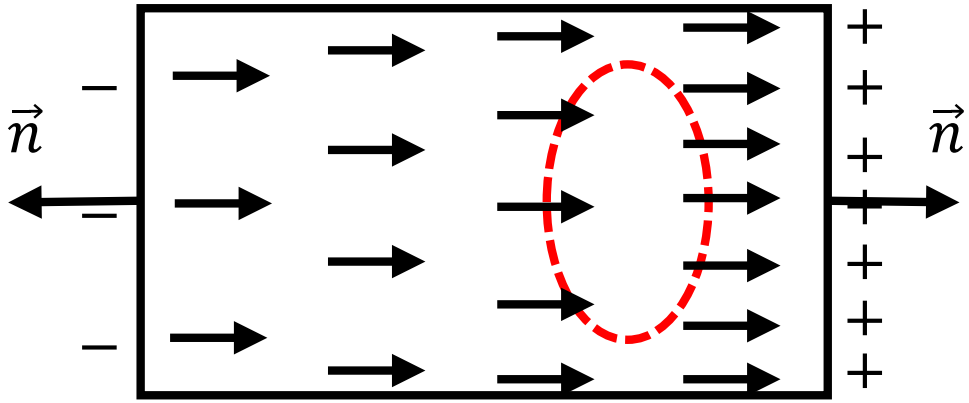
$\vec{E} = 0$

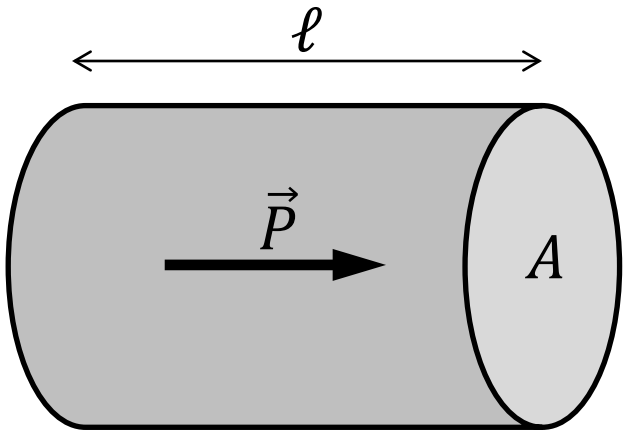


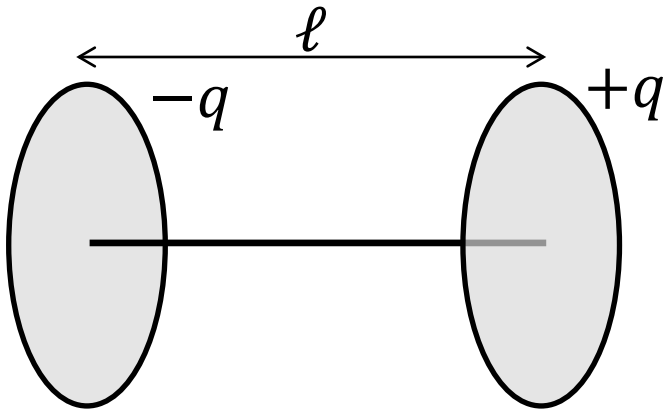
$\vec{E} \neq 0$

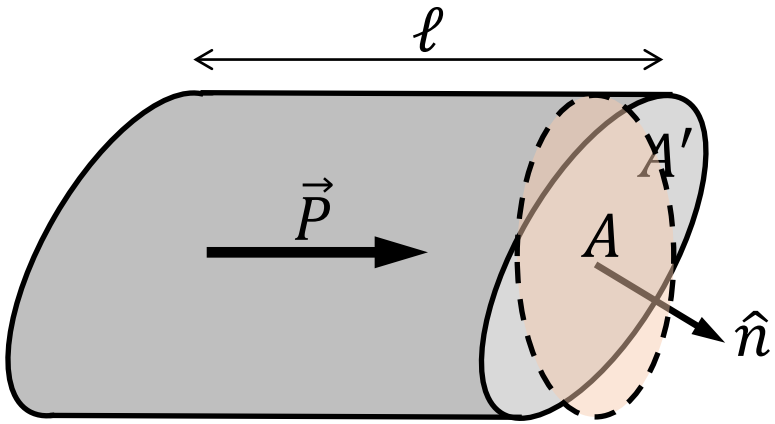


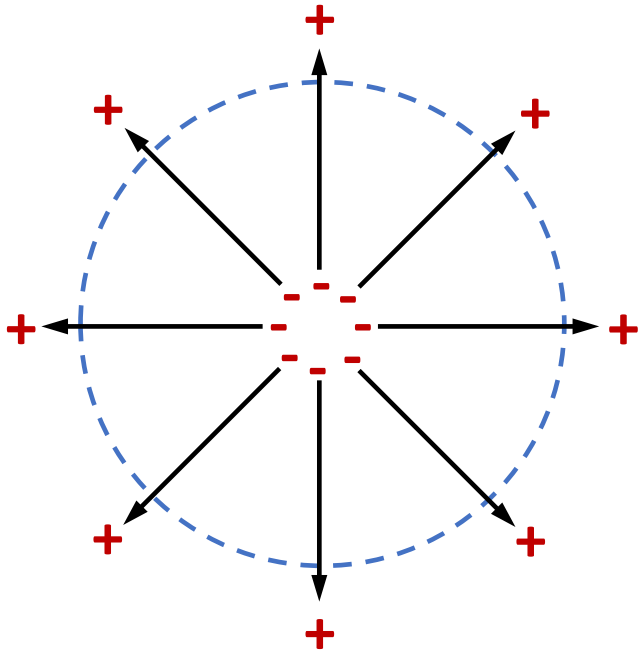

$\vec{n}$
$\vec{n}$


$\vec{n}$
$\vec{n}$

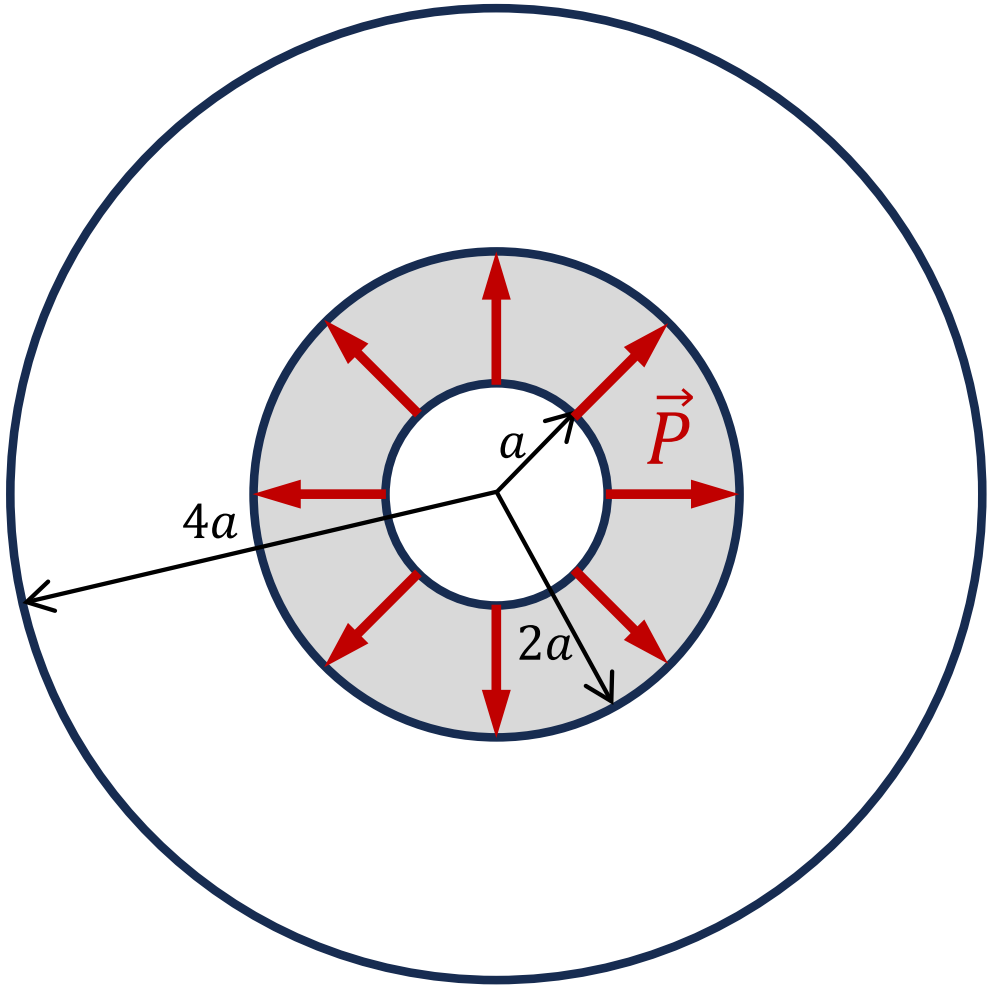

ℓ
$\vec{P}$
A


ℓ
$-q$
$+q$

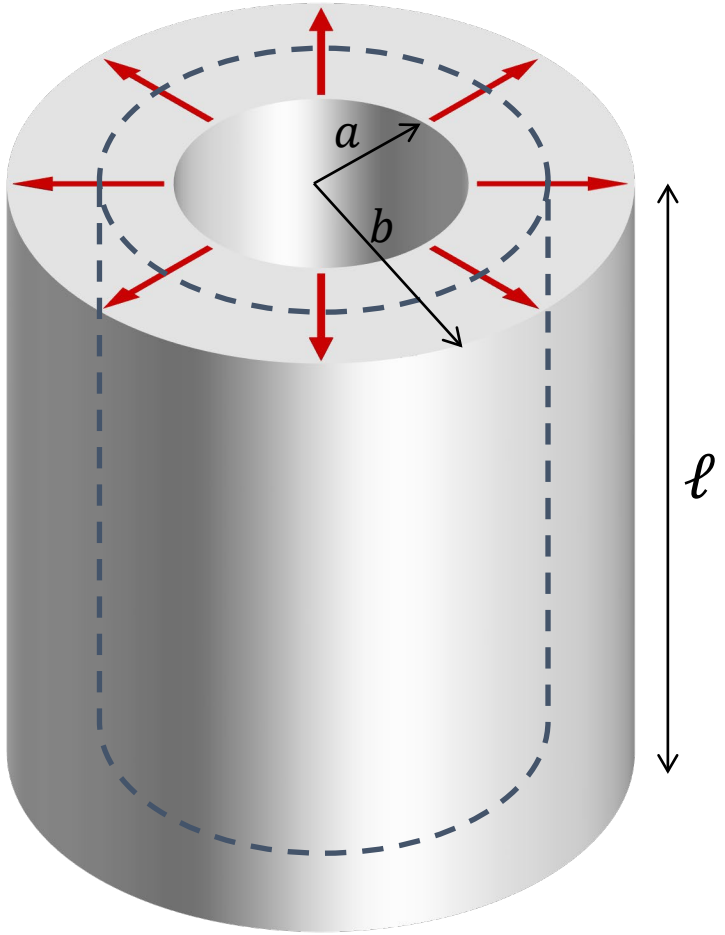

ℓ
$\vec{P}$
A'
A
$\hat{n}$

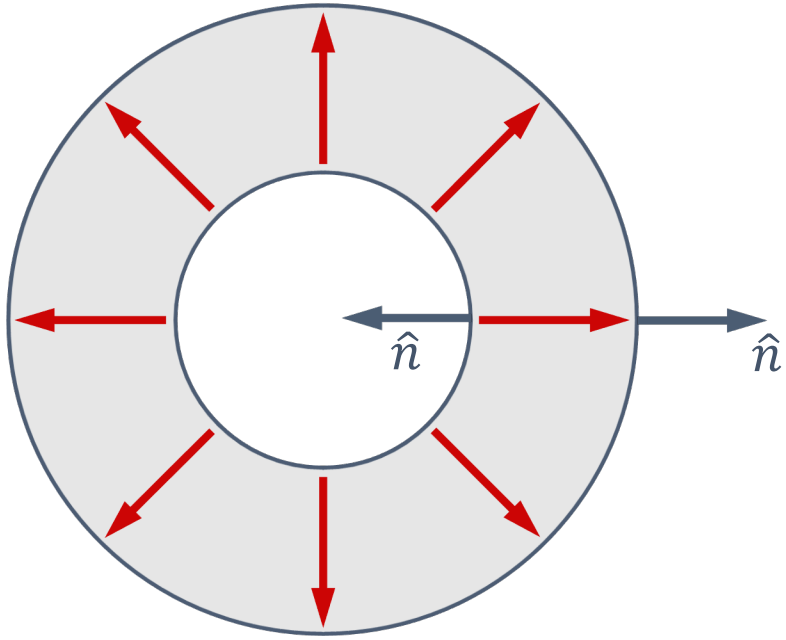


Problema 4.6

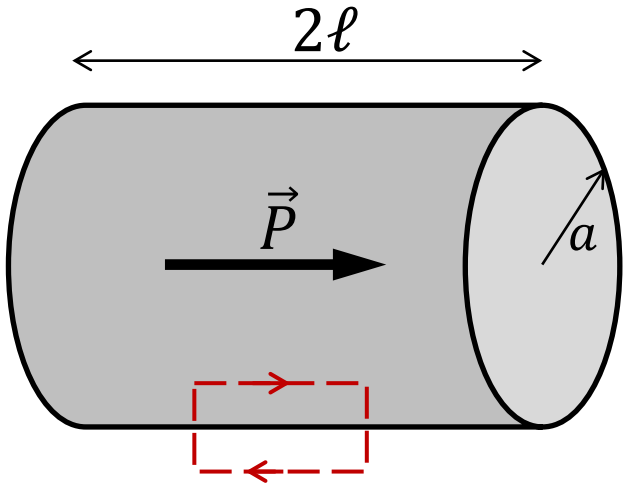


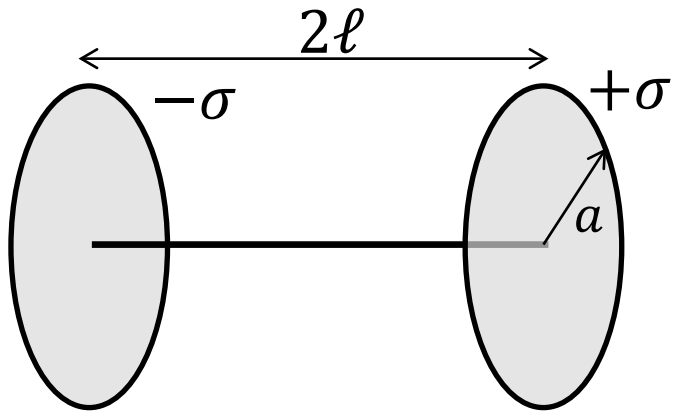
Problema 4.7



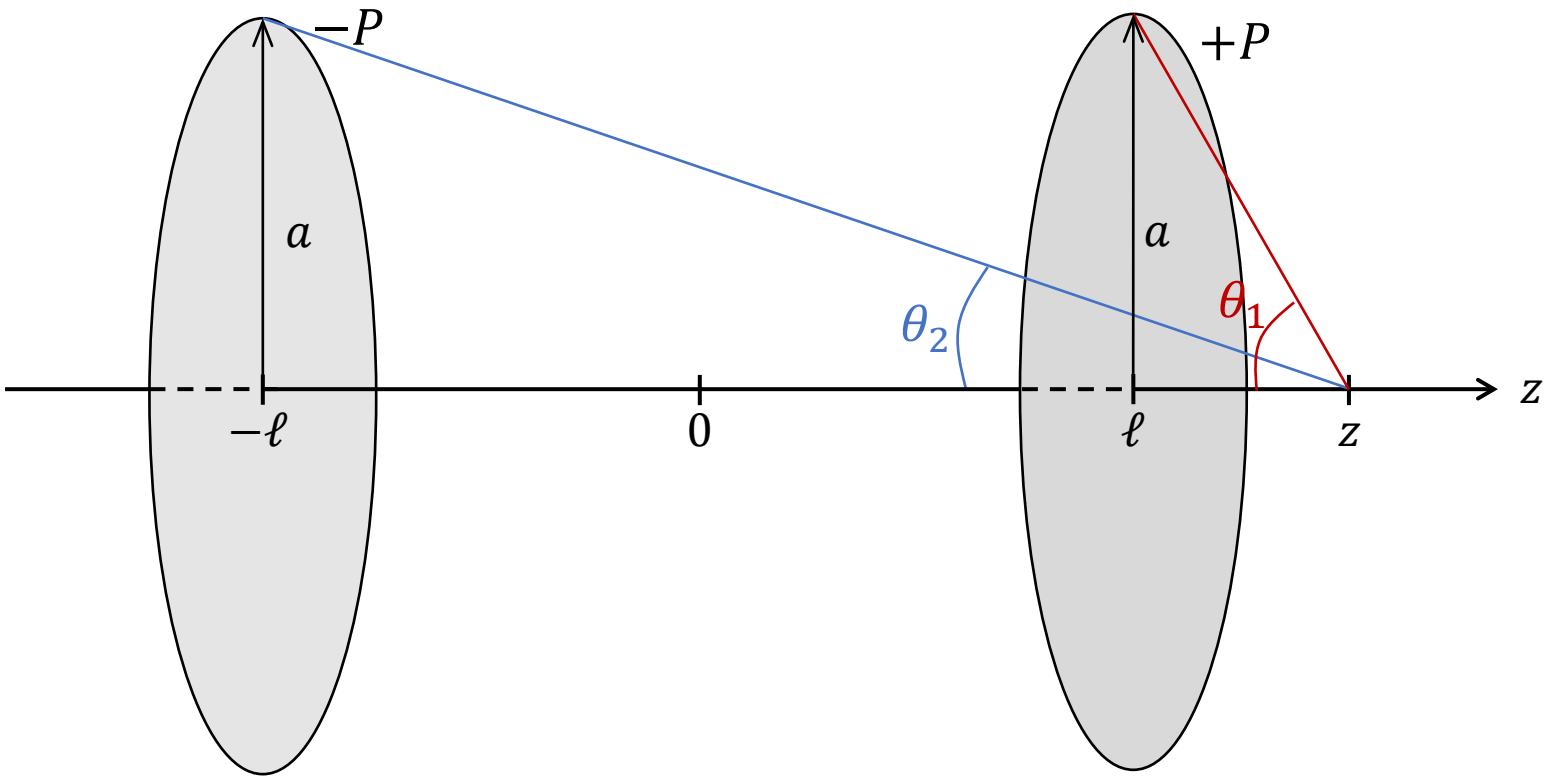


Problema 4.4

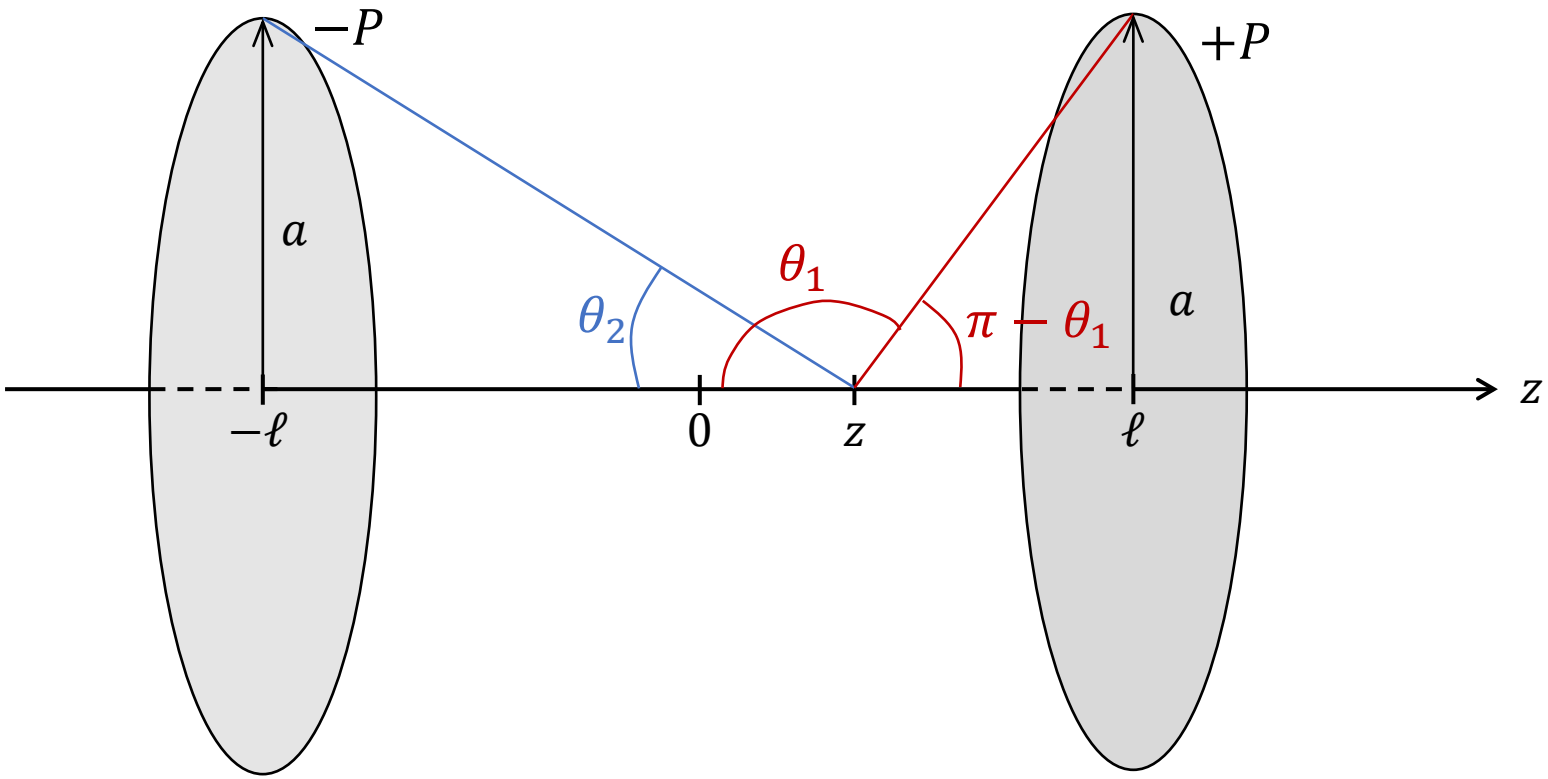


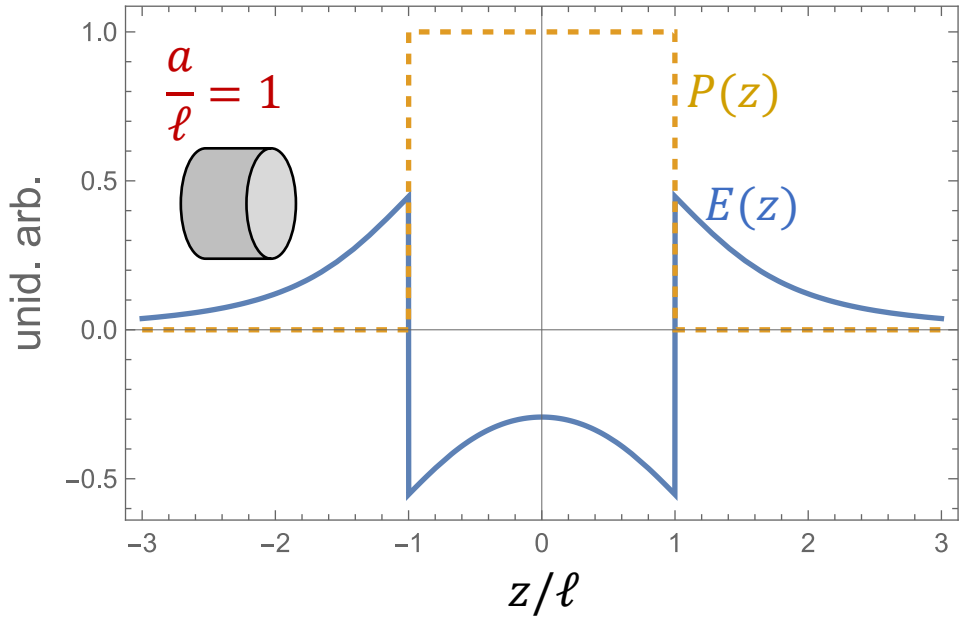


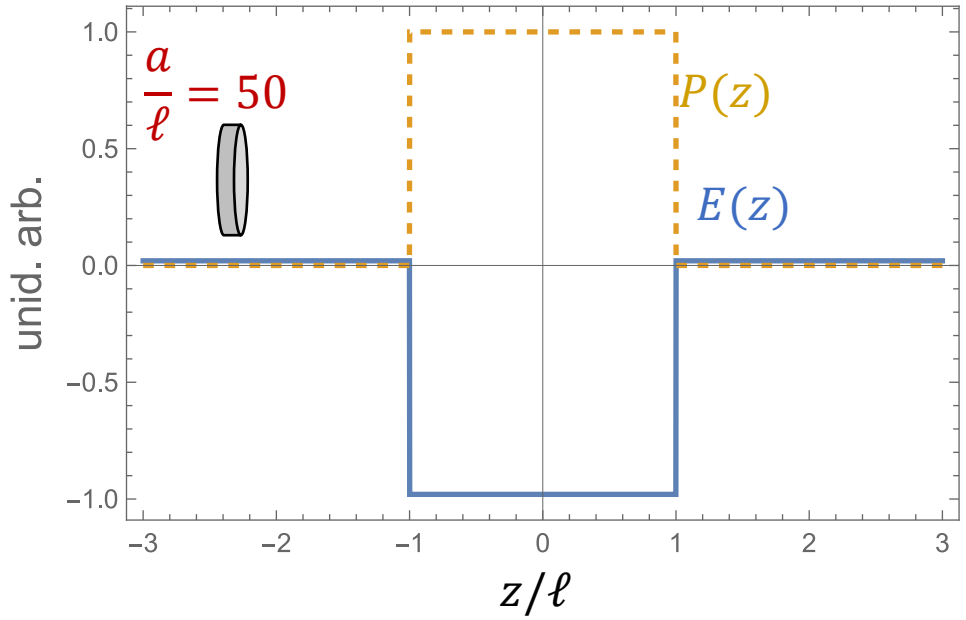
Problema 4.4

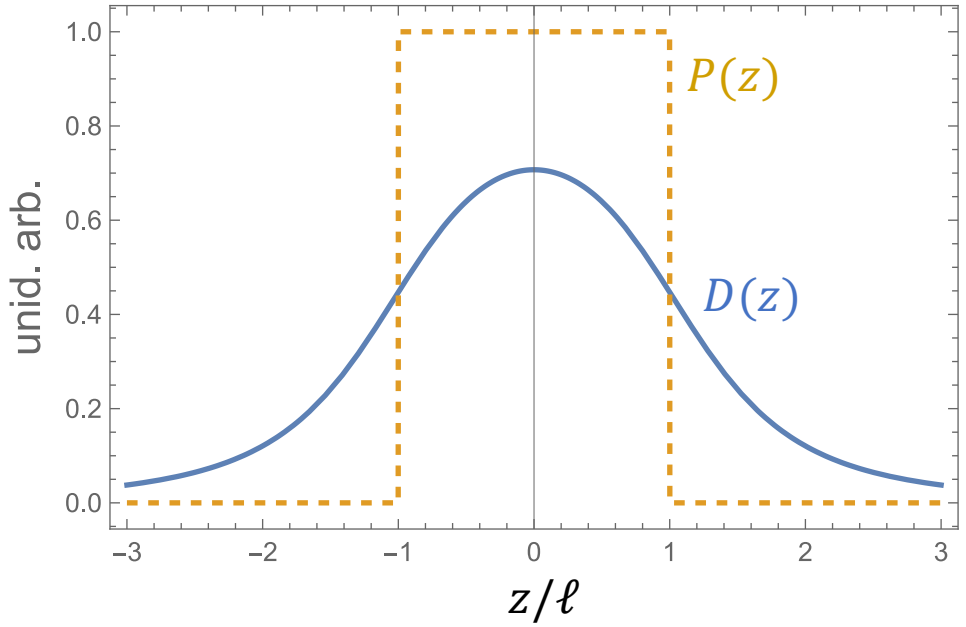


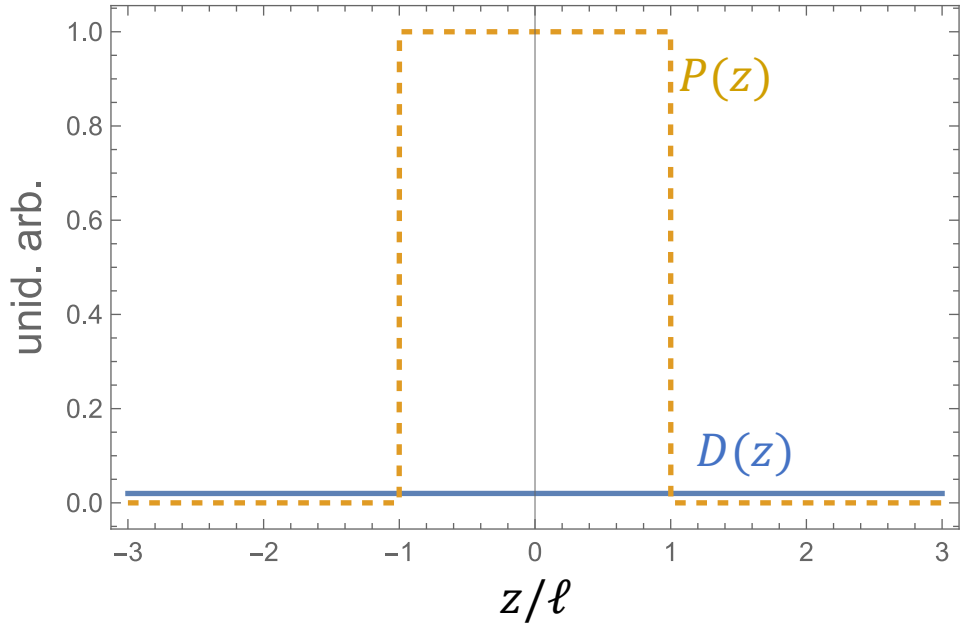
Problema 4.4

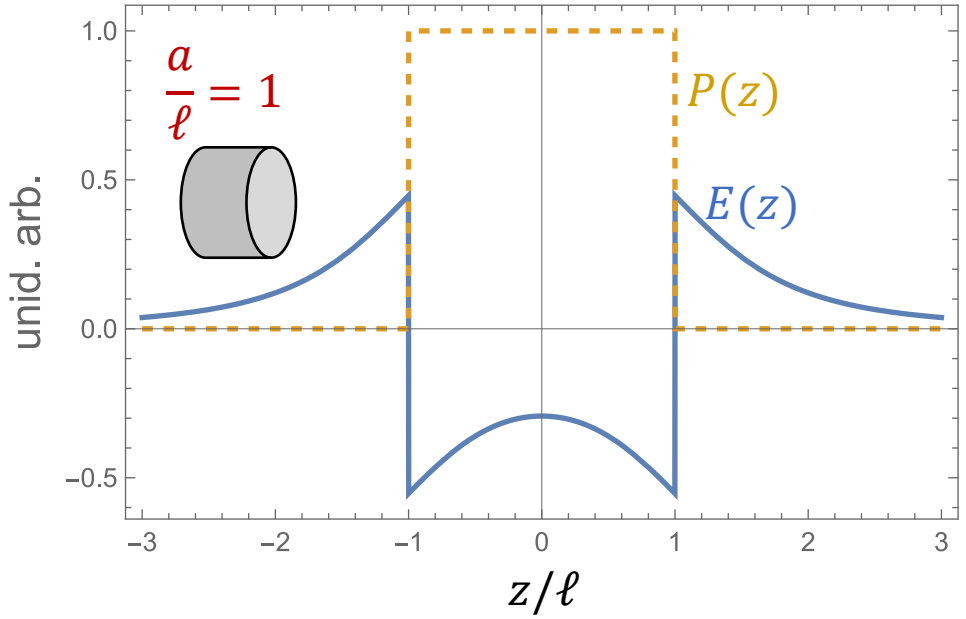


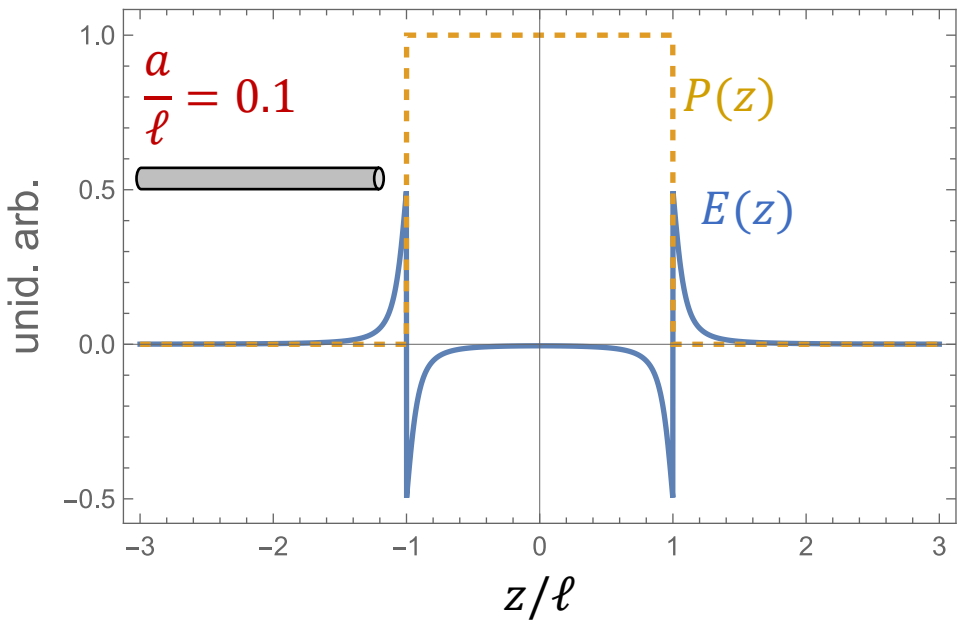

$\frac{a}{\ell} = 1$
$P(z)$
$E(z)$
unid. arb.
z/ℓ

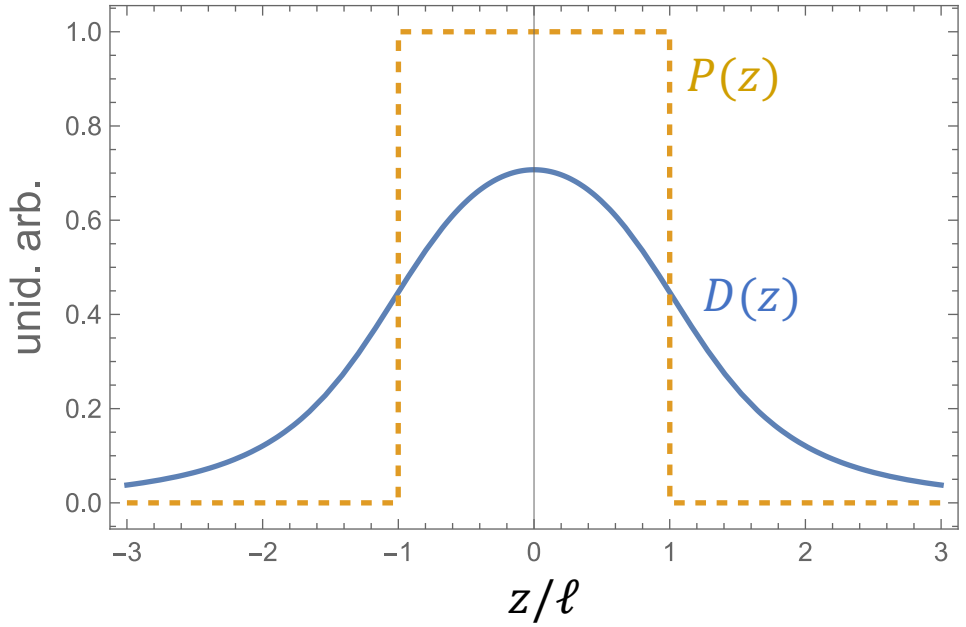

$\frac{a}{\ell} = 50$
$P(z)$
$E(z)$
unid. arb.
z/ℓ

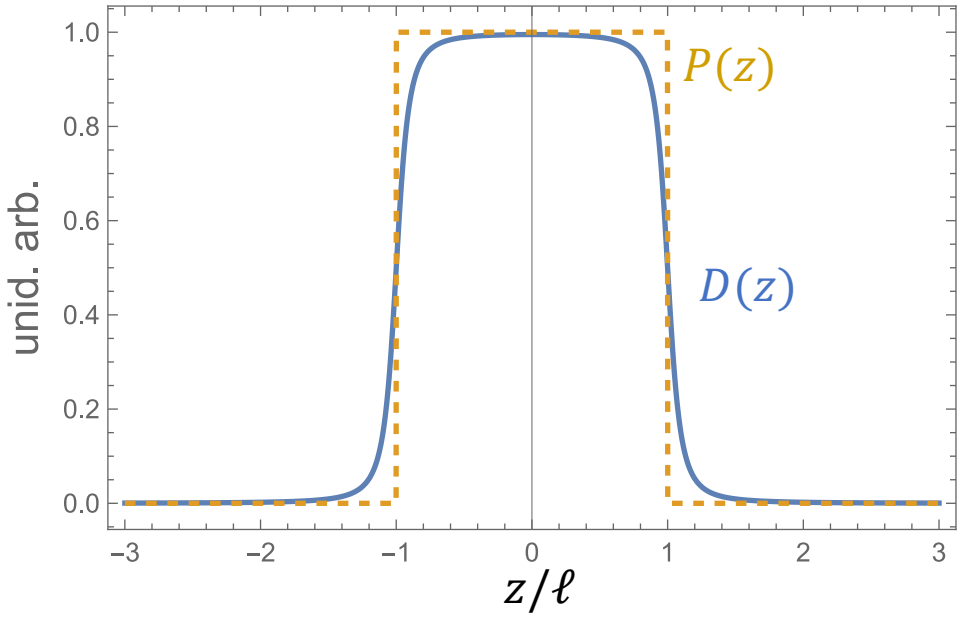

$P(z)$
$D(z)$
unid. arb.
z/ℓ


$P(z)$
$D(z)$
unid. arb.
z/ℓ

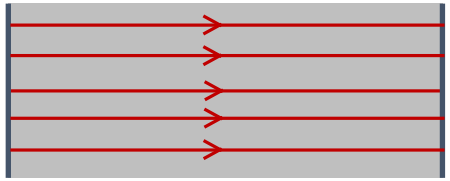

$\frac{a}{\ell} = 1$
$P(z)$
$E(z)$
unid. arb.
z/ℓ


$\frac{a}{\ell} = 0.1$
$P(z)$
$E(z)$
unid. arb.
z/ℓ

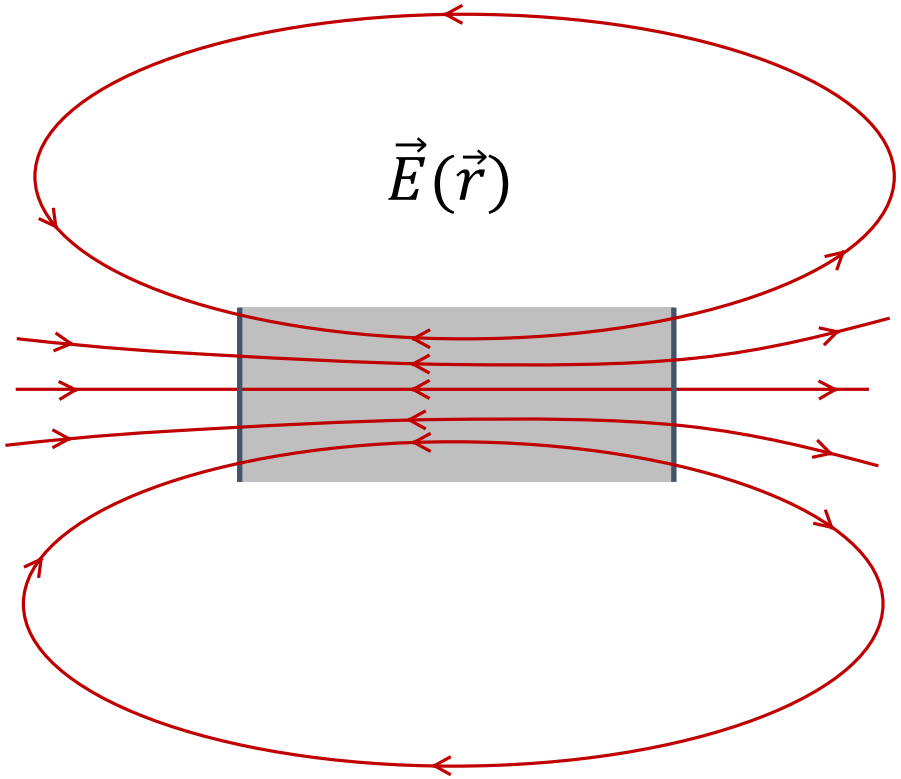

$P(z)$
$D(z)$
unid. arb.
z/ℓ


$P(z)$
$D(z)$
unid. arb.
z/ℓ

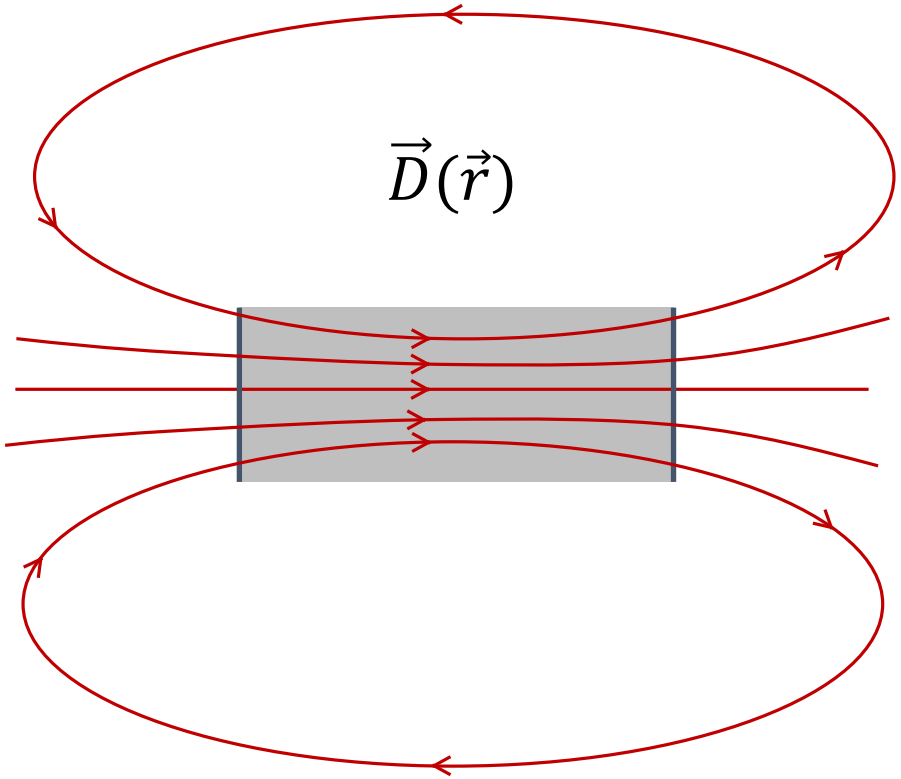
$\vec{P}(\vec{r})$



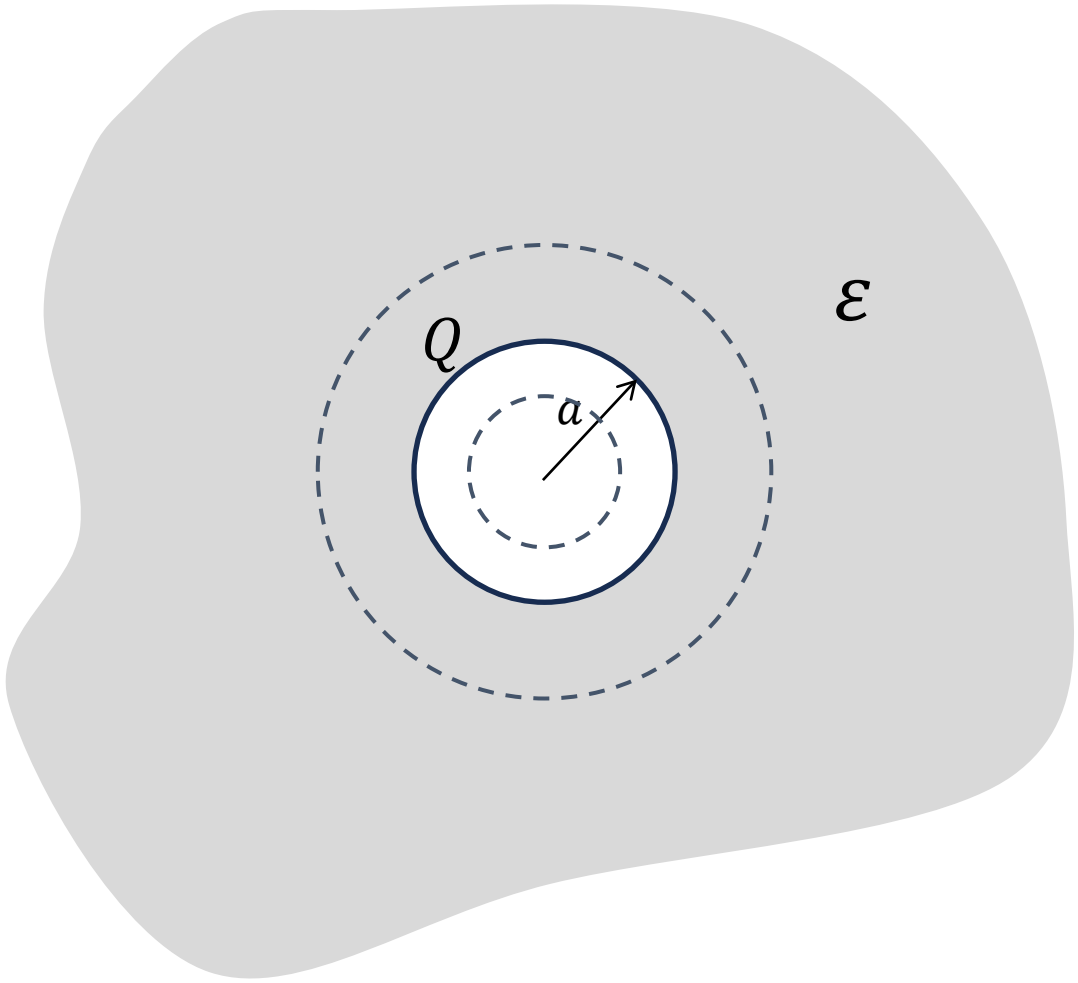
$\vec{E}(\vec{r})$



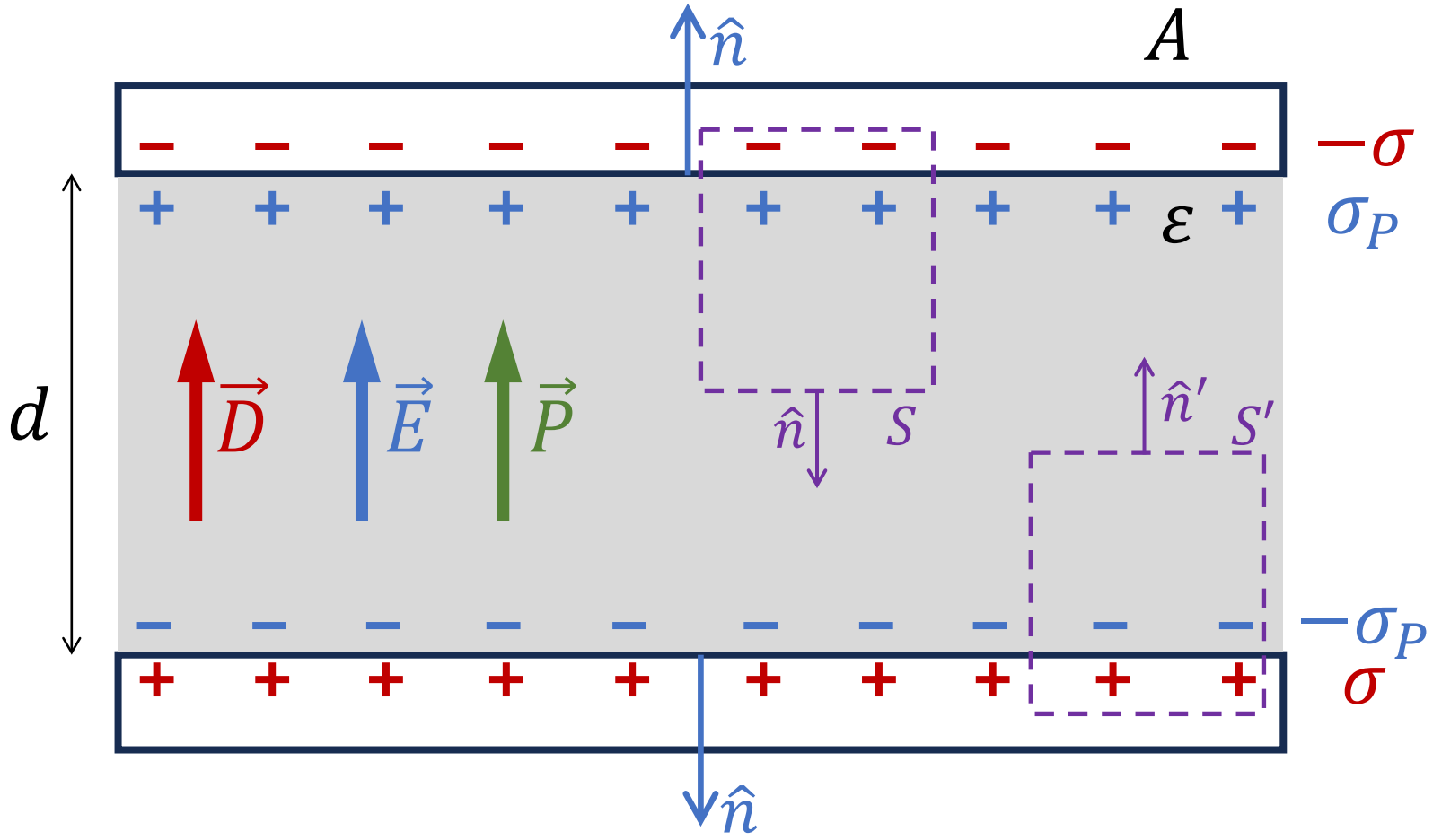
$\vec{D}(\vec{r})$

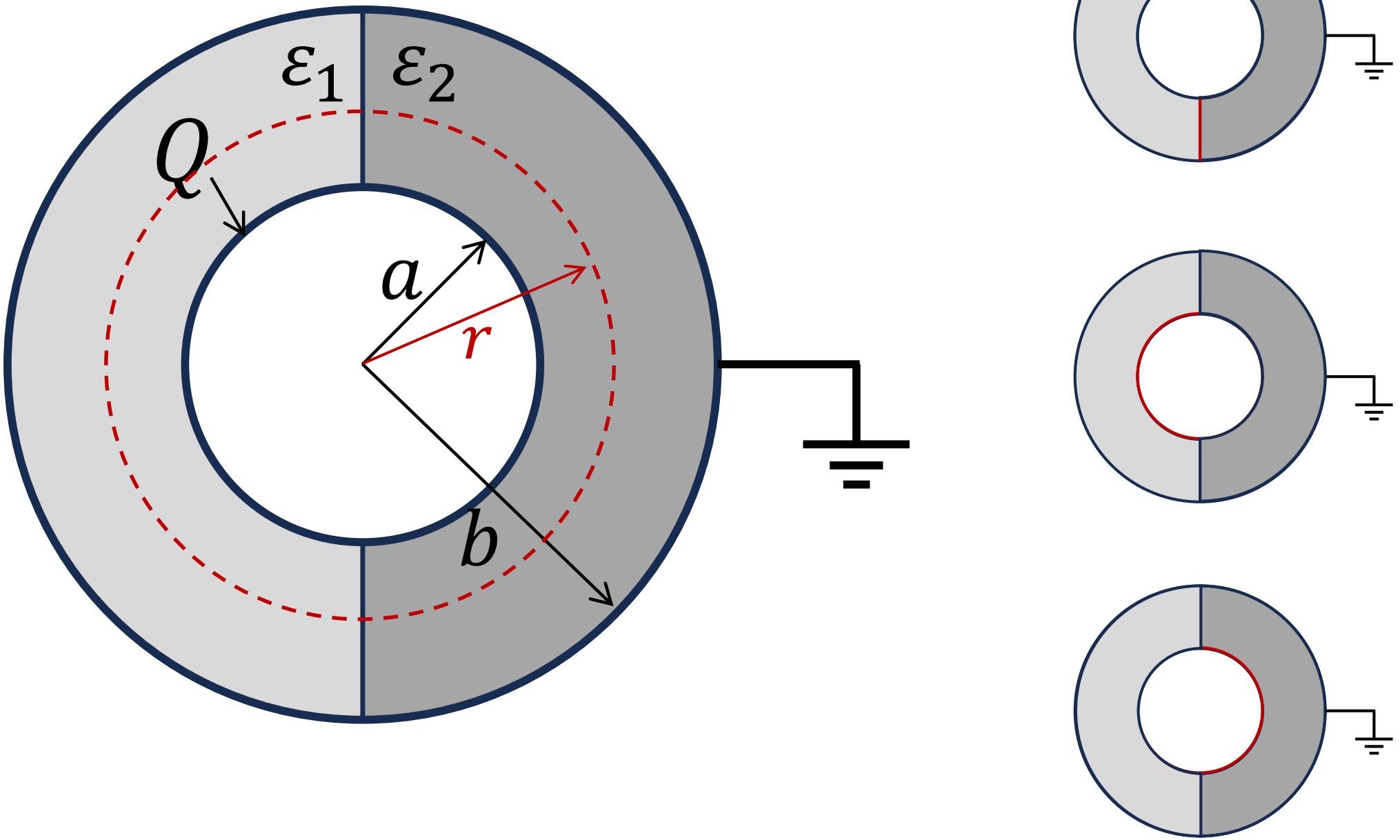


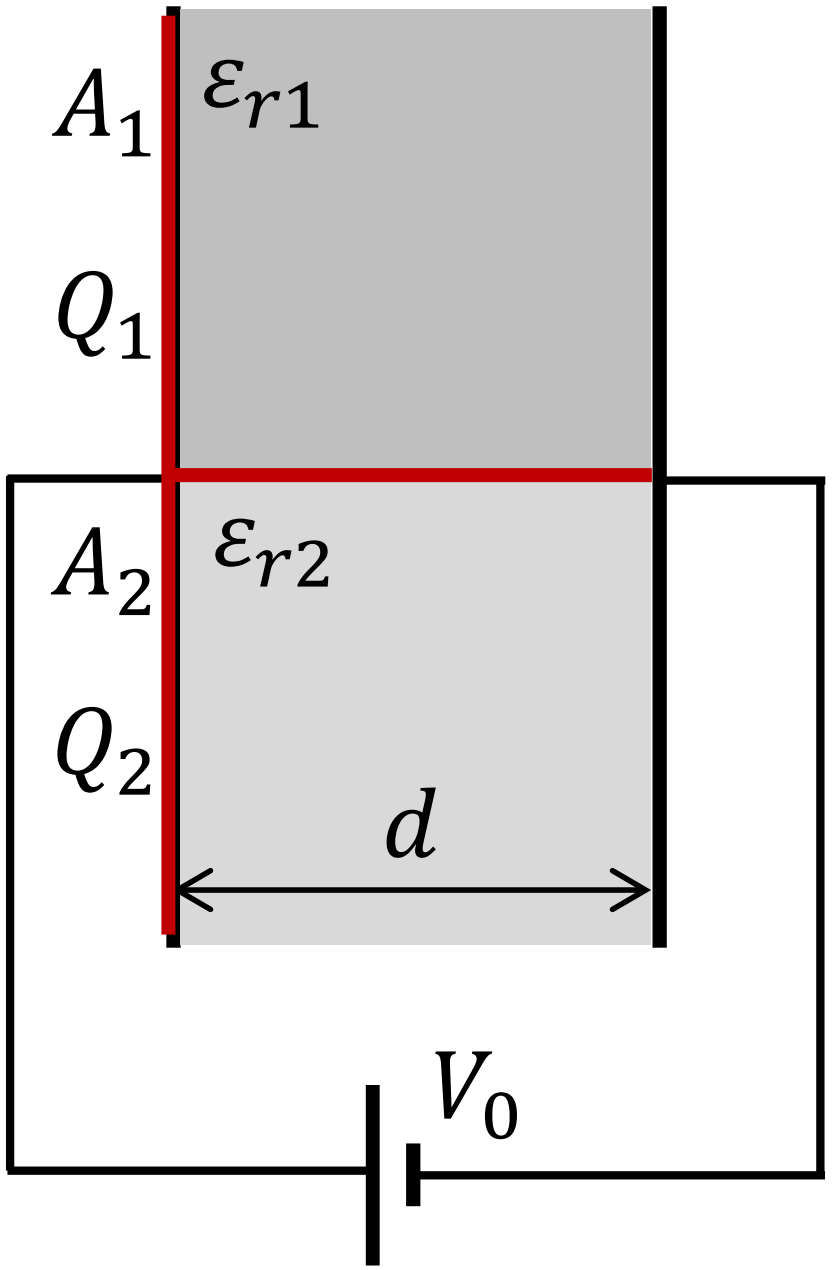
Problema 4.2

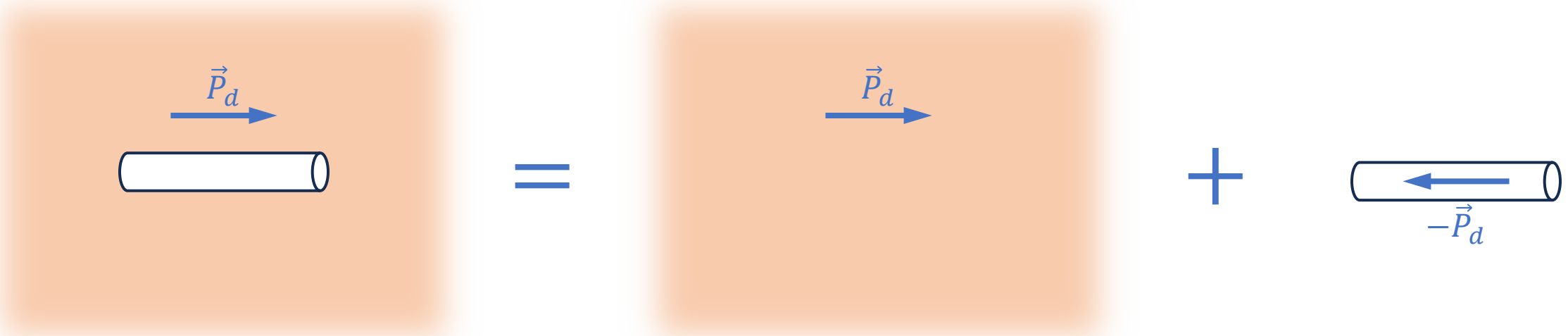


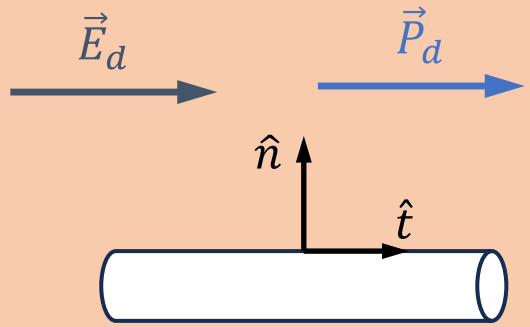
Problema 4.3


$\hat{n}$
A
$-\sigma$
σ_P
ε
d
$\vec{D}$
$\vec{E}$
$\vec{P}$
$\hat{n}$
S
$\hat{n}'$
S'
$-\sigma_P$
σ
$\hat{n}$

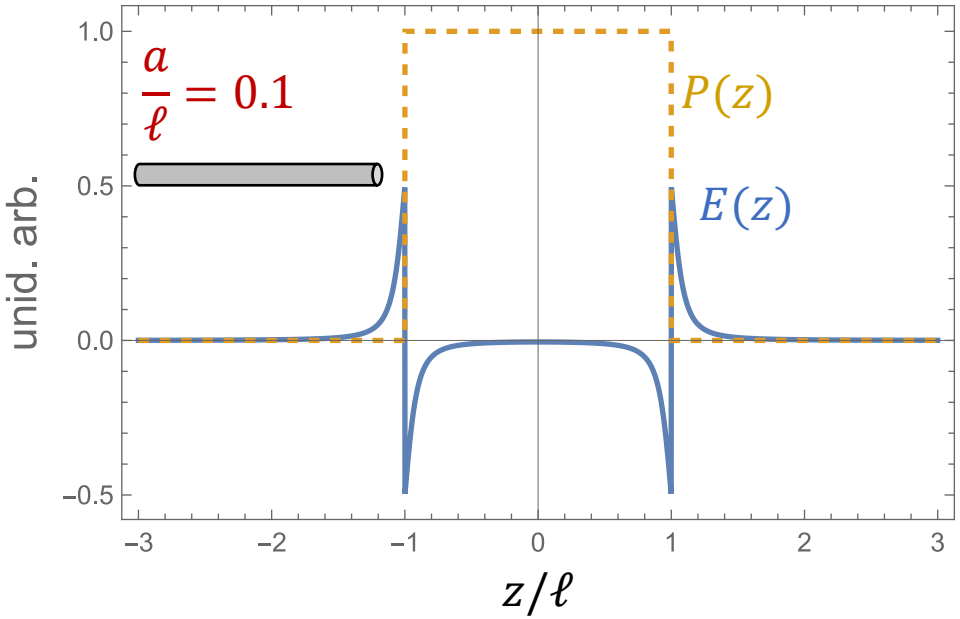

ε_1
ε_2
Q
a
r
b

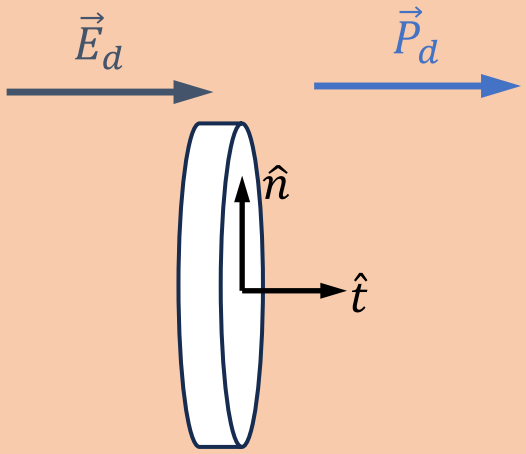

A_1
ε_{r1}
Q_1
A_2
ε_{r2}
Q_2
d
V_0


$\vec{P}_d$
=
$\vec{P}_d$
+
$-\vec{P}_d$

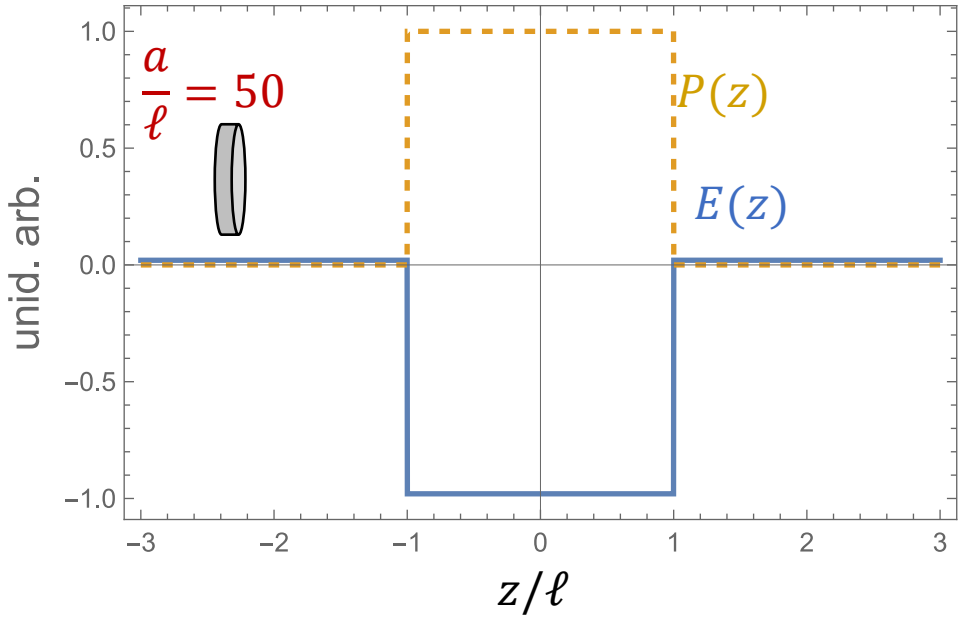

$\vec{E}_d$
$\vec{P}_d$
$\hat{n}$
$\hat{t}$

Problema 4.4


$\frac{a}{\ell} = 0.1$
$P(z)$
$E(z)$
unid. arb.
1.0
0.5
0.0
−0.5
−3
−2
−1
0
1
2
3
z/ℓ


$\vec{E}_d$
$\vec{P}_d$
$\hat{n}$
$\hat{t}$

Problema 4.4


$\frac{a}{\ell} = 50$
$P(z)$
$E(z)$
unid. arb.
1.0
0.5
0.0
−0.5
−1.0
−3
−2
−1
0
1
2
3
z/ℓ